

SP250 Homework 2

1. Let $n > 1$ be a number less than 10,000 that has no divisors greater than 1 and less than 100. Prove that n is a prime number.

Solution:

If n is composite, then n has divisors p and q such that $n = p \cdot q$ and p, q are both greater than 1. Since n has no divisors between 1 and 100, p and q must both be greater than or equal to 100. Hence, $n = p \cdot q \geq 100 \cdot 100 = 10,000 > n$, a contradiction. Thus n is not composite and must be a prime.

2. What is the sum of the first 100 numbers that are not divisible by 6?

Solution 1:

Among the first $6k$ positive integers, k are divisible by 6 and $5k$ are not divisible by 6. Hence, exactly 100 of the first 120 positive integers are not divisible by 6, namely 1, 2, 3, 4, 5, 7, 8, ..., 112, 113, 115, 116, 117, 118, 119. Using the rainbow method, the sum of these numbers is¹

$$(1 + 119) + (2 + 118) + (3 + 117) + \dots = 120 \cdot 100 / 2 = 6000$$

Solution 2:

We start the same way but use complementary counting to find the sum. That is, we sum

$$(1 + 2 + 3 + \dots + 119 + 120) - (6 + 12 + \dots + 120) = \frac{120 \times 121}{2} - 6 \times (1 + \dots + 20)$$

$$\text{This gives } 60 \times 121 + 6 \times \frac{20 \times 21}{2} = 60 \times 121 - 60 \times 21 = 60 \times 100 = 6,000.$$

3. Is it possible to connect 11 towns with roads in such a way that every town is connected to exactly 3 others?

Solution:

No, it is not possible. Say we keep track of the sum of the number of towns each town is connected to. When there are no roads at all, the sum is just zero. When a road is built between two towns, the sum increases by 2 since both towns are now connected to one more town. Hence, this sum will always be an even number. For each of the 11 towns to be connected to exactly 3 others, the sum will be 33, which is an odd number and therefore not possible.

¹ One has to be careful here since there are different gaps between the numbers in the sequence, but the gaps of two are symmetrical: we start with 5 consecutive numbers, then a gap of two, then 5 consecutive numbers, and so on – and there is the same pattern going back. This guarantees that all pairs sum up to the same number.

4. a) Find all prime numbers p such that $p^2 + 3$ is also a prime.
 b) Find all prime numbers p such that $p^2 + 2$ is also a prime.

Solution:

- a) Computing $p^2 + 3$ for the first few small prime p , we get

p	$p^2 + 3$
2	7
3	12
5	28
7	52
11	124
13	172

We see that $p^2 + 3$ is prime for $p = 2$. For all other values of p that we tried, $p^2 + 3$ is composite and divisible by 2. At this point, one may guess that $p^2 + 3$ is even and composite for all primes $p > 2$, but how do we prove it? To prove that $p^2 + 3$ is even and composite, we note that p^2 is odd for odd p , so $p^2 + 3$ is even for odd p . Since 2 is the only even prime, $p^2 + 3$ is composite for all odd p , and is only prime for $p = 2$.

- b) Computing $p^2 + 2$ for the first few small prime p , we get

p	$p^2 + 2$
2	6
3	11
5	27
7	51
11	123
13	171

One can make the following observations: (A) $p = 3$ works and (B) in all other cases, the result seems to be divisible by 3. We make a conjecture that (B) is always going to be true. Now we prove it.

Proof:

Any number is either divisible by 3, or is 1 more than a multiple of 3 (we will call it " $3k + 1$ for some non-negative integer k "), or is 2 more than a multiple of 3 (we will call it " $3k + 2$ for some non-negative integer k ").

If p is a multiple of 3, then $p = 3$ is the only possibility since p is prime. Checking, we find that $p = 3$ works.

If $p = 3k + 1$ then

$p^2 + 2 = (3k + 1)(3k + 1) + 2 = 9k^2 + 3k + 3k + 1 + 2 = 9k^2 + 6k + 3$, which is clearly divisible by 3 and is larger than 3, so it is composite.

If $p = 3k + 2$ then

$p^2 + 2 = (3k + 2)(3k + 2) + 2 = 9k^2 + 6k + 6k + 4 + 2 = 9k^2 + 12k + 6$, which is clearly divisible by 3 and is larger than 3, so it is composite.

Therefore, the only possibility is $p = 3$.

5. Joey has a gallon of wine and a gallon of water. He pours a pint of wine into water and then a pint of the mixture back into the container with wine. Does he now have more water in the wine container or more wine in the water container?

Solution:

Remarkably, it is the same. Suppose Joey has x gallons of water in the wine container. Because the total volume of the container is 1, there must be $1 - x$ gallons of wine in the wine container. Because the total volume of wine is 1 across the containers, there must be x gallons of wine in the water container, which proves our claim.

6. Two players play a game. The first player puts a chess rook anywhere on the board. After that, players alternate by moving the rook according to rules of chess, starting with the second player. It is forbidden to move the rook into a square that has already been visited, including the square where the rook was originally placed. The player who makes the final move wins. If both players play optimally, does one of the players have a winning strategy, and if so – which one?

Solution:

Player two has a winning strategy based on symmetry. Player two always mirrors player one's placement of the rook across the centerline of the chess board, making sure that the set of unvisited squares is always symmetrical with respect to the centerline after player two's move. This means that whichever unvisited square player one moves the rook to next, the mirror position on the other side of the centerline will also be unvisited, ensuring that player two is always able to move the rook to the mirror position as well as preserve the symmetry of the unvisited squares.